

## Accuracy

# An introduction to model accuracy and metrics

# What is a model?

A model is a **mathematical representation** of a real-world process or **relationship between variables**.

For example, the **line of best fit is a linear model** used to represent the relationship between two variables.

**Real-world situations often involve more complex relationships, requiring more detailed models.**

Models often have multiple variables and interactions to better capture the details of the processes they represent.

# What is model accuracy?

**Model accuracy** refers to how well a **model is able to accurately predict values**. We use various **metrics** to **measure model accuracy**.

## The importance of model accuracy

- Ensuring that our model is **accurate** helps us **make better predictions and informed decisions**.
- An **inaccurate model** can lead to **incorrect conclusions and poor decision-making**.
- Evaluating model accuracy helps us **identify areas for improvement and refine our model**.

# Residuals

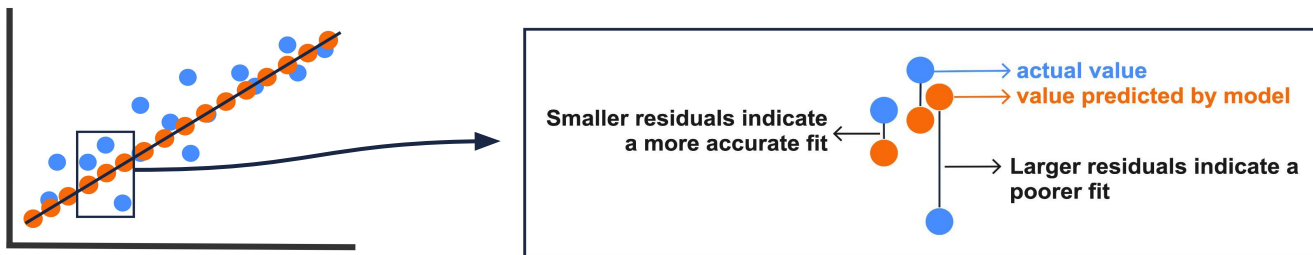
A residual is the **difference between** the **observed value** and the **predicted value** for a data point. They are also referred to as **errors**.

Residuals help us understand **how well our model is fitting the data**.

By analysing the residuals, we can **identify patterns and trends** that our model may not be capturing.

## When to use them?

- Residuals are useful for understanding the **pattern of errors** the model is making and **diagnosing issues** with the model.
- They can also be used to **identify outliers** or **influential data points**.



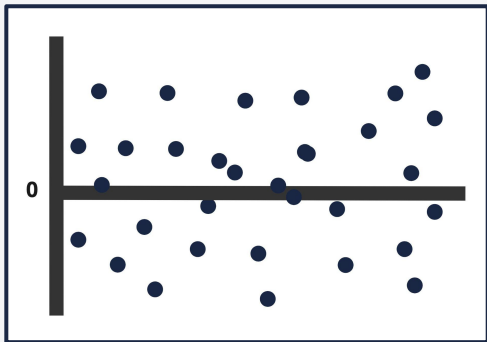
# Residual plot

Residual plots are **scatter plots of residual values**. They help determine the **accuracy** of a model.

## What a residual plot should look like

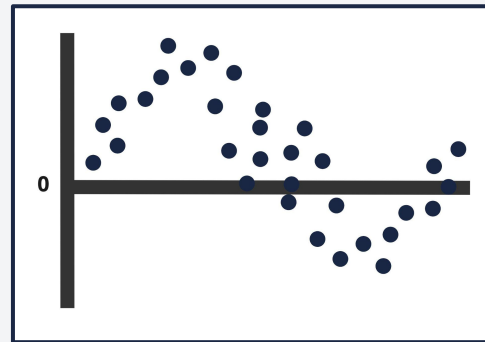


Residuals are **randomly scattered around zero**, with **no discernible pattern**.



## What a residual plot shouldn't look like

**Clear patterns or trends** in residuals suggest bias or shortcomings in the model.



# MAE, MSE, and RMSE

Three commonly used measures of accuracy are MAE, MSE, and RMSE. These measures **quantify the errors or residuals** of a model's predictions and provide a way to **compare the accuracy of different models**.

## Mean Absolute Error (MAE)

The **average of the absolute values of the residuals**. It represents the average distance between the predicted values and the actual observed values.

## Mean Squared Error (MSE)

The **average of the squared residuals**. It emphasises larger errors by squaring them, making it more sensitive to outliers.

## Root Mean Squared Error (RMSE)

The **square root of the MSE**. It represents the average distance between the predicted values and the actual observed values, similar to MAE, but with more emphasis on larger errors.

# Understanding MAE for model accuracy

The **average of the absolute values of the residuals**.

$$\text{MAE} = \frac{\sum_{i=1}^n |y_i - x_i|}{n}$$

where  $y_i$  is the predicted value,  $x_i$  is the observed value, and  $n$  is the number of data points.

## Analysis

- ✓ It gives an idea of how **far the predictions are**, on average, **from the true values**.
- ✓ A **lower** MAE value indicates a **more accurate** model. For example, if the MAE is 2, it means that, on average, the model's predictions are off by 2 units.
- ✓ MAE is **easy to interpret** and **less sensitive to outliers**.

# Understanding MSE for model accuracy

The **average of the squared residuals**.

$$\text{MSE} = \frac{\sum_{i=1}^n (y_i - x_i)^2}{n}$$

where  $y_i$  is the predicted value,  $x_i$  is the observed value, and  $n$  is the number of data points.

## Analysis

- ✓ It is the **average of the squared differences** between the predicted values and the actual values of the target variable.
- ✓ A **lower** MSE value indicates a more **accurate** model. For example, if the MSE is 4, it means that, on average, the model's predictions are off by 4 **squared** units.
- ✓ MSE **penalises larger errors** more heavily than smaller errors as errors are squared and, therefore, larger errors have a greater impact on the overall value.



# Understanding RMSE for model accuracy

The **square root** of the MSE.

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (y_i - x_i)^2}{n}}$$

where  $y_i$  is the predicted value,  $x_i$  is the observed value, and  $n$  is the number of data points.

## Analysis

- ✓ It represents the **average distance between the predicted values and the actual observed values**, similar to MAE, but with more **emphasis on larger errors**.
- ✓ A **lower** value of RMSE indicates a more accurate model.
- ✓ RMSE is more interpretable than MSE, as it is in **the same unit as the original data**. For example, if the RMSE is 2, it means that, on average, the model's predictions are off by 2 units, in the same units as the target variable.

# Interpreting accuracy in context

To **determine the accuracy** of our model, it's crucial to **interpret metrics within the context** of the scenario we are investigating.

**Question:** We have calculated an MAE of 4 units. **Is our model accurate?** Let's consider two scenarios:

## Predicting a student's final grade

If we aim to **predict a student's final grade**, an **MAE of 4 units** indicates our model is relatively **accurate**. This is because we consider a **predicted mark of 80%** quite close to an actual mark of 84%.

## Predicting shoe size

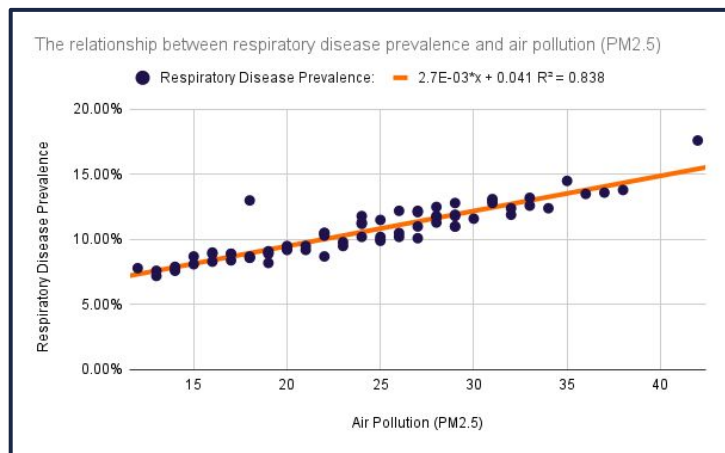
If we aim to **predict the shoe size** someone will buy, an **MAE of 4 units** indicates high **inaccuracy**. This is because a shoe size difference of 4 units, such as a size 5 to size 9, is significantly different.



The same logic applies to **residuals**, **MSE**, and **RMSE**. Therefore, it is important to always **consider the context** of the problem when interpreting our results.

# Calculating MAE, MSE, and RMSE – high accuracy

Let's calculate MAE, MSE, and RMSE to assess the **accuracy** of a **linear model** (the line of best fit) representing the relationship between **air pollution and respiratory disease prevalence** in African cities.



**MAE = 1.3308**

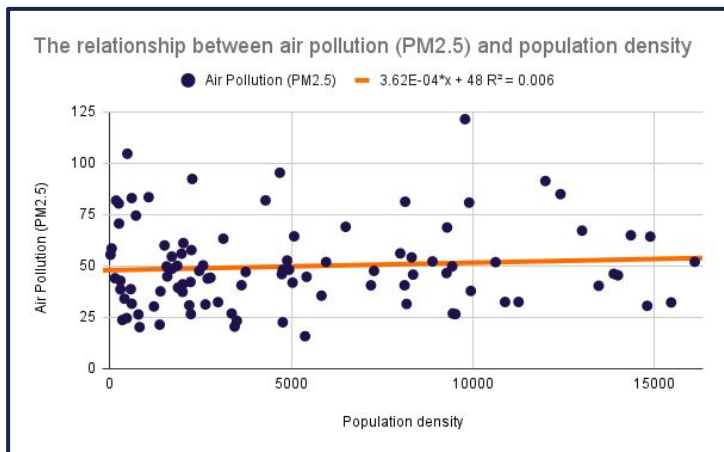
**MSE = 2.2809**

**RMSE = 1.5104**

- An **MAE of 1.3308** suggests, on average, the **predicted** respiratory disease prevalence values **deviate from the actual values by 1.33 percentage points**. The relatively **low MAE** suggests that the linear model provides reasonably **accurate predictions**.
- The **MSE of 2.2809** suggests that there is **some variability** in the predicted values compared to the actual values, although it is **small**.
- The **RMSE of 1.5104** indicates that, on average, the predicted respiratory disease prevalence values deviate from the **actual values by approximately 1.51 percentage points**. The low value indicates our model is making **accurate predictions**.

# Calculating MAE, MSE, and RMSE – low accuracy

Let's calculate MAE, MSE, and RMSE to assess the **accuracy** of a **linear model** (the line of best fit) representing the relationship between **population density** and **air pollution levels** in African cities.



$$\text{MAE} = 15.7641$$

$$\text{MSE} = 419.9802$$

$$\text{RMSE} = 20.4934$$

- An **MAE of 15.76** suggests that, on average, the **predicted** air pollution values **deviate from the actual values by 15.76 units**. The relatively **high MAE** suggests that the model's predictions have a moderate level of error and may **not be accurate**.
- An **MSE of 419.98** is relatively **high**. This suggests there are **considerable differences** between the values, indicating the model's predictions may **not be accurate**.
- An **RMSE of 20.49** is **high**. This suggests that the model's predictions have a **moderate to high level of deviation** from the actual values.

# Comparison of accuracy metrics

## Residuals

By examining the residuals, we can assess if there are any **patterns or trends** in the model's **predictions**.

We can examine the **residual plot** to assess the **accuracy** of the model.

## MAE

**Easy to interpret** and **less sensitive to outliers**, which means it provides a good measure of average prediction error.

However, it **doesn't punish large errors** more than small ones. This can be a drawback when large errors are considered critical.

## MSE

**Punishes large errors** more than small errors because the errors are squared. This is useful when we want to emphasise large errors.

However, MSE is **more sensitive to outliers** and it is **not as simple to interpret**.

## RMSE

Combines the **advantages of both MAE and MSE**.

It **punishes large errors** more than small ones, similar to MSE. It is also more **interpretable as it is in the same units as the target variable**.

However, it remains **sensitive to outliers**.